BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions (MCQ's)

> Note: Q.1 Choose the correct answer for each from the given options:

Q1. A simple covalent molecule possesses two bond pairs and two lone pairs around the central atom, its shape should be:

(a) linear
(b) Planar Trigonal
(c) angular
(d) None

Q2. The correct relation between debye and coulomb meter is:

(a) 1D = 3.33 x 10^-30 Cm
(b) 1D = 1.6 x 10^-19 Cm
(c) 1D = 1.88 x 10^-12 Cm
(d) 1D = 1.23 x 10^-9 Cm

Q3. The bond order of N2 molecule is:

(a) 0
(b) 1
(c) 2
(d) 3

Q4. For Avogadro's number, this statement is incorrect:

(a) It is the no. of particles in one moles of any substances.
(b) Its numberical value is 6.02 x 10^23
(c) Its value change if temperature increases
(d) Its value change if number of moles increases

Q5. The minimum number of moles are present in:

(a) 1 dm3 of methane gase at STP
(b) 5 dm^3 of helium gas at STP
(c) 10 dm3 of hydrogen a gas at STP
(d) 22.4 dm^3 of chlorine gas at STP

Q6. In 1935 A.D James Chadwick was awarded Nobel Prize because...

(a) He Discovered proton
(b) He discovered neutron
(c) He determined the radius of hydrogen atom
(d) He gave the rule for electronic configuration

Q7. When 4d orbital is filled, the next electron enter into

(a) 5s
(b) 5p
(c) 5d
(d) 6s

Q8. According to Graham's Law of diffusion, the ratio of diffusion of H2 and O2 are respectively.

(a) 1:2
(b) 2:1
(c) 1:4
(d) 4:1

Q9. Collection of gas over water is an example of:

(a) Graham's law
(b) Dalton's
(c) Avogadro's law
(d) Gaylusac law

Q10. Under similar condition CH4 diffuses ..... Times faster than SO2 gas:

(a) 1.5 time
(b) 2 time
(c) 4 time
(d) 16 time

Q11. The boiling point of water (H2O) is 100°C where as that of hydrogen sulphide (H2S) is 420°C, This can be attributed to:

(a) Smaller bond angle of H2S than H2O
(b) Smaller radii of oxygen than sulphur
(c) High I.P of oxygen than sulphur
(d) Tendency of water to form hydrogen bond

Q12. Liquids can form convex meniscus in a narrow glass tube when:

(a) Cohesive forces are stronger than adhesive forces
(b) Adhesive forces are stronger than cohesive forces
(c) Cohesive and adhesive forces are equal in strength
(d) None of these

Q13. The equilibrium of $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ (ΔH = +ve) is affected by change in:

(a) Temperature only
(b) Pressure only
(c) Both and pressure
(d) Neither temperature nor pressure

Q14. In reaction $A(g) + B(g) \rightleftharpoons 2C(g)$, the equilibrium constants Kp = Kc because:

(a) $\Delta n > 1$
(b) $\Delta n < 1$
(c) $\Delta n = 1$
(d) $\Delta n = 0$

Q15. The term Elive mass use in law of mass action means:

(a) No. of mole
(b) No. of molecules
(c) mole dm3
(d) gram per dm3

Q16. In the equilibrium system of PCl5(g) ⇌ PCl3(g) + Cl2(g), the correct relationship between Kc and Kp is:

(a) Kp>Kc
(b) Kp<Kc
(c) Kp=Kc
(d) Kp/Kc = 1

Q17. The equilibrium of which of the following reaction would not be affected by an increase in pressure:

(a) PCl5(g) ⇌ PCl3(g) + Cl2(g)
(b) 2NO(g) + Cl2(g) ⇌ 2NOCl(g)
(c) N2(g)+O2(g) ⇌ 2NO(g)
(d) 2SO2(g) O2(g) 2NO3(g)

Q18. The unit of rate constant (K) for the first order reaction is:

(a) s^-1
(b) conc. s^-1
(c) conc^-1. s
(d) conc.^-1 s^-1

Q19. Rate constant of a reaction is affected by:

(a) Conc. of reactants
(b) Conc. of products
(c) Temperature
(d) reaction time

Q20. Identify the incorrect statement about colloidal solution:

(a) It shows Tyndall effect
(b) Its particles movement is Brownian type
(c) Its particle size is less than 1 nm
(d) Its physical appearance is translucent

Q21. For a hypothetical reaction x + y → z, if the conc. of x is double, the rate increases by square and if the conc. of y is doubled the rate increases by twice. The experimental rate law of this reaction is:

(a) R = K [x]^1 [y]^1
(b) R = K [x]^1 [y]^2
(c) R = K [x]^2 [x]^1
(d) R = K [x]^2 [y]^2

Q22. The unit of rate constant (K) for the first order reaction is:

(a) s^-1
(b) conc. s^-1
(c) conc^-1. s
(d) conc.^-1 s^-1

Q23. Rate constant of a reaction is affected by:

(a) Conc. of reactants
(b) Conc. of products
(c) Temperature
(d) reaction time

Section-B (Short Questions)

> Note: Answer any 09 questions of the following:

Q2. Define rounding of data give various rules of rounding fo data.

Q3. What is gram molecular mass?

Q4. Explain why the filling of electrons in as orbital takes place prior 3d?

Q5. Write down the electronic configuration of the following.: (i) Fe (Z=26) (ii) Br- (z=35) (iii) Ca2+(z=20) (ii) Br- (z=35)

Q6. The Dipole moment of water is 1.85 D but CO2 has zero dipole mement why?

Q7. Define Isotropy. Which type of solids have this property?

Q8. What is transition temperature? Explain with examples.

Q9. What is Salt? Explain acidic, Basic and neutral salts?

Q10. Name various units of concentration and explain mole fraction?

Q11. What is transition temperature? Explain with examples.

Q12. Give three properties of each α, β and γ rays.

Q13. What is meant by diffusion and effusion?

Section-C (Descriptive Question)

> Note: Answer any 03 question of the following:

Q14. (a) Define theoretical yield, actual yield and percent yield. why the practical yield is often less (b) Amonia gas can be produced by heating together the solid NH4Cl and Ca(OH)2: 2NH4Cl + Ca(OH)2 → 2NH3 + CaCl2 + 2H2O

Q15. Silver Sulphide (Ag2S) is an anti microbial. In experiment 24.8g Ag2S is reacted with the excess of hydrochloric acid [illegible in source scan]

Q16. Describe hydrogen bonding iin water and explain the anomalous behavior of water due to hydrogen bonding? (b) The pH of solution 25°C is 4.82, calculate its hydrogen ion concentration.

Q17. what is surface tension, Explain it with example. Give its unint and describe the factors that affect on surface tension of liquids?

Q18. How can ideal gas is differentiated from real gas? What are the causes of deviation of real gas from ideal behavior? Explain theis deviation at low temperature and high pressure?